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Online terrain estimation for autonomous vehicles on deformable terrains

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Abstract

In this work, a terrain estimation framework is developed for autonomous vehicles operating on deformable terrains. Previous work in this area usually relies on steady state tire operation, linearized classical terramechanics models, or on computationally expensive algorithms that are not suitable for real-time estimation. To address these shortcomings, this work develops a reduced-order nonlinear terramechanics model as a surrogate of the Soil Contact Model (SCM) through extending a state-of-the-art Bekker model to account for additional dynamic effects. It is shown that this reduced-order surrogate model is able to accurately replicate the forces predicted by the SCM while reducing the computation cost by an order of magnitude. This surrogate model is then utilized in a unscented Kalman filter to estimate the sinkage exponent. Simulations suggest this parameter can be estimated within 4% of its true value for clay and sandy loam terrains. It is also shown in simulation and experiment that utilizing this estimated parameter can reduce the prediction errors of the future vehicle states by orders of magnitude, which could assist with achieving more robust model-predictive autonomous navigation strategies.

Keywords: Terramechanics, parameter estimation, wheeled vehicles, deformable terrain, control, Kalman Filter

1. Introduction

Autonomous ground vehicles (AGVs) have drawn interest for military applications to perform tasks, such as supply transport, in unsafe environments that could pose a threat to human operators (Iagnemma and Dubowsky, 2002). Three considerations about military AGVs are important to motivate this work. First, military vehicles often need to operate off-road on deformable terrains, where the vehicle’s mobility is dependent on the highly nonlinear tire forces generated at the tire-terrain interface (Taheri et al., 2015). Second, increasing the mobility of military AGVs is a critical need (Liu et al., 2017). Third, state-of-the-art approaches to navigate such vehicles typically rely on model dependent architectures, such as Model Predictive Control (MPC) (Liu et al., 2017, 2018). Therefore, when the AGVs are operated on deformable terrains, a more accurate knowledge of the terrain parameters becomes a critical enabler to maximize the mobility of the AGVs.

Much research has been performed in developing terramechanics models for off-road applications, which can be divided into empirical models, physics-based models, semi-empirical models (Taheri et al., 2015). Empirical models are the simplest; however, such models do not generalize well beyond the experimental test conditions used for their development. On the other hand, physics-based finite and discrete element models have proven to be of the highest fidelity, but the large computational efforts required renders them infeasible for real-time tire force prediction, thus limiting their applicability for use in AGVs and real-time terrain estimation (Taheri et al., 2015). More promising candidates, and perhaps the most widely used, for real-time tire force prediction on deformable terrains are the semi-empirical models based upon the classical terramechanics theory developed by Bekker, including the Soil Contact Model (Gallina et al.,

2014; Ishigami et al., 2007; Smith, 2014; Guo, 2016). In these models, the tire is typically assumed rigid and the deformation is assumed to occur only in the terrain (Smith, 2014). To model the complex tire-terrain interactions, these terramechanics models rely on knowledge of terrain-specific parameters such as cohesion, internal friction angle, or sinkage exponent. During vehicle operation, these parameters may not be explicitly known or may be varying due to non-uniform terrains. Therefore, real-time terrain estimation is necessary in AGVs to improve the accuracy of the terramechanics models online and generate better informed control commands. Having this capability would also provide insight into traversability of terrains, such that path planning algorithms can reroute the vehicle to avoid regions where loss of mobility or excessive power consumption is likely to occur (Iagnemma, 2006).

Researchers have already recognized this need and a limited number of results are available in the literature. In particular, in (Gallina et al., 2014, 2016), a Bayesian procedure is utilized for terrain parameter identification, but making this approach work online is subject to future research. Other researchers have proposed an online algorithm for estimating soil cohesion and internal friction angle utilizing a linear least-squares estimator for a rover (Iagnemma and Dubowsky, 2002; Iagnemma et al., 2004). The algorithm relies on simplifying classical terramechanics equations through linear approximations to increase computational efficiency and subjects the rover to periodic high and low speed traverses (Iagnemma and Dubowsky, 2002). Similarly, in (Ding et al., 2015), multiple terrain parameters are estimated through least squares curve fitting with promising results for a sand-like terrain. Due to large computation times in the Bekker-based model, on the order of 10 s to estimate the parameter, the model is linearized. Multiple linearized models are analyzed which take on the order of 15 ms to hundreds of ms depending on model simplicity, sac-

87 rificing estimation accuracy for computational efficiency. In (Hutangkabodee
 88 et al., 2006), the Composite Simpson’s Rule and Newton Raphson method are
 89 utilized for online determination of terrain parameters from measured forces.
 90 In (Ray, 2009), a multi-model approach is used to efficiently select a terrain
 91 parameter set from a precompiled library of parameter sets and mappings de-
 92 termined *a priori*. Finally, in (Setterfield and Ellery, 2013) polynomial models
 93 representing terrain response are developed from measurements including rover
 94 velocities and forces rather than explicitly estimating the terrain parameters,
 95 but real-time implementation was not achieved.

96 While existing results are promising, several limitations of the existing liter-
 97 ature are important for this work: (1) linear approximations can lead to inaccu-
 98 rate stress approximations, particularly in terrains with low sinkage exponents
 99 such as clay (Zhenzhong Jia et al., 2011), and hence inaccurate force prediction,
 100 (2) these algorithms often rely on measurements obtained from force and torque
 101 sensors, which are not necessarily standard on AGVs, and (3) periodically oper-
 102 ating at low speeds is not desirable when maximum mobility is desired. Hence,
 103 online estimation of deformable terrain parameters for off-road AGVs is still an
 104 open research area and is the focus of this work.

105 This study presents a new approach for online terrain parameter estimation.
 106 First, due to the large computation time associated with integrating stresses in
 107 SCM and limitations of classical terramechanics equations, a nonlinear reduced-
 108 order model is developed by extending the work presented in (Zhenzhong Jia
 109 et al., 2011) to account for additional dynamic effects such that a sufficient
 110 agreement with SCM can be achieved. Then, the reduced-order terramechanics
 111 model is incorporated in a 3 DoF bicycle model (Liu et al., 2018) to create
 112 an estimation model, whereas the actual vehicle is represented with a 11 DoF
 113 plant model with SCM or a physical vehicle operating on a grass field. The

114 predictions from the estimation model are fused with measurements from the
 115 plant model, independent of forces and torque measurement, in an Unscented
 116 Kalman Filter (UKF) to identify the dominant terrain parameter, namely, the
 117 sinkage exponent. The result is an online terrain estimation approach that can
 118 be used to better inform control and path-planning algorithms for AGVs.

119 The rest of this paper is organized as follows. Sec. 2 first briefly reviews
 120 the SCM model used in the plant model. Then a state-of-the-art fast terrame-
 121 chanics model used as a benchmark is introduced and the significant deviations
 122 of its predictions from SCM are demonstrated. This model is then modified
 123 to improve its accuracy vis-à-vis SCM, so that a suitable estimation model is
 124 obtained. Sec. 3 presents the vehicle models, both the plant model as well
 125 as the estimation model. The terrain estimation procedure based on UKF is
 126 summarized in Sec. 4. Sec. 5 and Sec 6 give the simulation and experimen-
 127 tal conditions. Sec. 7 gives the simulation results, including the accuracy of
 128 the estimations and their ability to improve the predictive accuracy of the 3
 129 DoF model, and the experimental results on grass. Finally, Sec. 8 gives the
 130 conclusions drawn from this work.

131 2. Terramechanics Models

132 2.1. Soil Contact Model (SCM)

133 This section briefly reviews the terramechanics model adopted in this work to
 134 represent the tire-terrain interactions in the plant simulations with high fidelity.
 135 This model is also used to evaluate the accuracy of the fast terramechanics
 136 models, including a state-of-the-art model and the surrogate model developed
 137 in this work. As such, this model serves as the ground truth for the purposes of
 138 this work.

139 The terramechanics model used in this study for generating the lateral tire

forces acting on the vehicle is based on the Soil Contact Model (SCM) reported in (Gallina et al., 2014; Krenn and Gibbesch, 2011). Verification of the model can be found in (Krenn and Hirzinger, 2009). The SCM calculates relevant forces and torques acting on a 3 dimensional object in contact with a deformable terrain as summarized below.

The SCM algorithm relies on a discretized mesh of the tire and terrain to search for contact points at the tire-terrain interface. In the contact detection step, the vertices of the tire mesh are projected onto the nearest vertices of the terrain digital elevation map, effectively arranging the contact vertices in individual columns. The sinkage at each vertex can be determined from the minima of each column, assuming the vertex location is a point of sinkage. The effective contact width, b , can then be determined from the footprint's area and contour length (Krenn and Gibbesch, 2011).

Following contact detection, the algorithm calculates the stresses at each contact node of the footprint as follows. The pressure, σ , is expressed as (Bekker, 1962)

$$\sigma = (k_c/b + k_\phi)h^n \quad (1)$$

The shear stress, τ , is expressed as (Janosi et al., 1961)

$$\tau = \tau_{\max}(1 - e^{-j/k}) \quad (2)$$

with $\tau_{\max}$ given as

$$\tau_{\max} = (c + \sigma \tan \phi) \quad (3)$$

In the above expressions h is the sinkage, b is the tire effective width, and j is the shear deformation. The remaining parameters are internal parameters characterizing the terrain as summarized in Table 1. The forces generated at the tire-terrain interface can then be given by integrating the stresses over the

Table 1: SCM terrain parameters.

Parameter	Symbol	Unit
Cohesive modulus	k_c	N/m ^{$n+1$}
Frictional modulus	k_ϕ	N/m ^{$n+2$}
Sinkage exponent	n	-
Shear deformation modulus	k	m
Cohesion	c	Pa
Angle of internal friction	ϕ	rad

entire contact patch. The above overview is a summary of (Gallina et al., 2014); a more complete discussion is given in (Krenn and Gibbesch, 2011).

SCM is a rather complex model due to the discretizations and integrations involved and may thus not be suitable for real-time parameter identification purposes. It has been shown that the accuracy of SCM is heavily influenced by the discretization resolution (Krenn and Gibbesch, 2011). Furthermore, several SCM operations are of N^2 complexity, where N is the number of grid nodes (Krenn and Gibbesch, 2011). As an example, for a discretization of just 200 total nodes (100 per tire in a bicycle model), the time required by SCM can be around 20 ms (Krenn and Gibbesch, 2011). Furthermore, for the UKF, the estimation method used in this work, 17 sigma points must be generated as discussed in Sec. 4, each calling the terramechanics model twice (once per tire in the bicycle model). Thus the total time spent calculating tire forces can be around 350 ms per a single UKF iteration. Finally, taking into account that many UKF iterations are needed to achieve estimation convergence, a UKF with SCM can be expected to take several minutes to converge, which is impractically long. Therefore, faster terramechanics models are needed.

2.2. State-of-the-Art Fast Terramechanics Model

Much less computationally demanding solutions better suited for online estimation are given by Bekker-based models. These models are again based on

182 (1)-(3); however, σ , τ , h , and, j are now replaced by functions of the angle of
 183 contact, θ . As such, (1)-(3) are rewritten as:

$$\sigma(\theta) = (k_c/b + k_\phi)h(\theta)^n \quad (4)$$

$$\tau(\theta) = \tau_{\max}(1 - e^{-j(\theta)/k}) \quad (5)$$

$$\tau_{\max} = (c + \sigma(\theta) \tan \phi) \quad (6)$$

186 where $h(\theta)$ is given by

$$h(\theta) = \begin{cases} r(\cos \theta - \cos \theta_f) & \theta_m \leq \theta \leq \theta_f \\ r(\cos \theta_e - \cos \theta_f) & \theta_r \leq \theta \leq \theta_m \end{cases} \quad (7)$$

187 with

$$\theta_m = (a_0 + a_1 s)\theta_f \quad (8)$$

$$\theta_f = \cos^{-1}(1 - h_f/r) \quad (9)$$

$$\theta_e = \theta_f - (\theta - \theta_r)(\theta_f - \theta_m)/(\theta_m - \theta_r) \quad (10)$$

$$\theta_r = \cos^{-1}(1 - \Lambda h/r) \quad (11)$$

191 where r is the radius of the tire; θ_f is the angle at which the front of the tire
 192 comes into contact with the terrain; θ_m is the location of maximum normal
 193 stress with a_0 and a_1 as terrain parameters typically taking on values of 0.4 and
 194 0–0.3, respectively (Wong and Reece, 1967); s is the longitudinal slip of the tire;
 195 θ_e is the equivalent front contact angle for angles less than θ_m ; θ_r is the angle at
 196 which the rear of the tire loses contact with the terrain; and Λ is a property of
 197 the terrain characterizing the sinkage ratio. Note that Λh in (11) corresponds
 198 to the h_r in Fig. 1a.

199 Finally, $j(\theta)$ is given as

$$j(\theta) = \begin{cases} r[(\theta_f - \theta) - (1 - s)(\sin \theta_f - \sin \theta)] & s \geq 0 \\ r[(\theta_f - \theta) - (1/(1 + s))(\sin \theta_f - \sin \theta)] & s < 0 \end{cases} \quad (12)$$

200 The maximum sinkage can be calculated in an iterative fashion by using the
 201 Newton-Raphson method as proposed in (Guo, 2016) as follows. The maximum
 202 sinkage is initialized as the static sinkage, which is based on the load on the tire
 203 W :

$$h_0 = \left[\frac{3W}{b(3 - n)(k_c/b + k_\phi)\sqrt{2r}} \right]^{\frac{2}{2n+1}} \quad (13)$$

204 However, due to dynamic effects, such as slippage, additional sinkage is induced.
 205 To account for this, the reaction force is calculated as

$$F_z = \int_{\theta_r}^{\theta_f} rb(\tau(\theta) \sin(\theta) + \sigma(\theta) \cos(\theta))d\theta \quad (14)$$

206 and a new sinkage is determined using the Newton-Raphson root finding method
 207 as

$$h'_0 = h_0 - F_z(h_0)/F'_z(h_0) \quad (15)$$

208 where F'_z denotes the derivative of the reaction force with respect to sinkage.
 209 The iterative procedure terminates when the calculated reaction force is within
 210 a specified tolerance of the normal force applied to the tire. Once the maximum
 211 sinkage is determined, the lateral force F_y can then be calculated in a similar
 212 fashion as in (Ishigami et al., 2007; Guo, 2016), i.e.,

$$F_y = \int_{\theta_r}^{\theta_f} rb\tau_y(\theta) \quad (16)$$

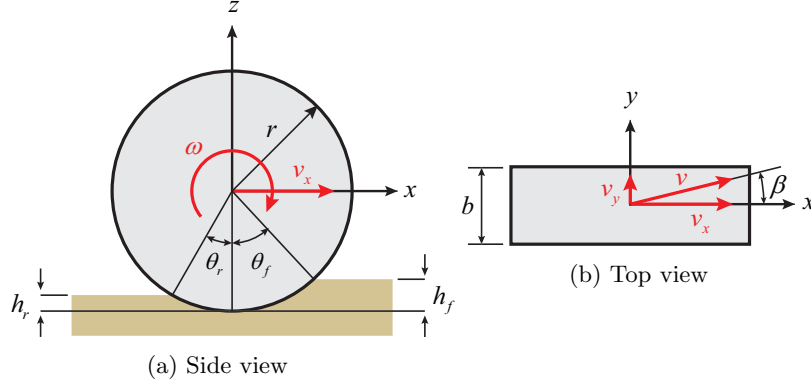


Figure 1: Tire-terrain geometry for positive slip.

213 with $\tau_y(\theta)$ given as

$$\tau_y(\theta) = \tau_{\max}(1 - e^{-|j_y(\theta)|/k_y}) \quad (17)$$

214 where

$$j_y(\theta) = r(1 - s)(\theta_f - \theta) \tan \beta \quad (18)$$

215 and β is the side slip angle given as

$$\beta = \arctan\left(\frac{v_y}{v_x}\right) \quad (19)$$

216 Depending on the soil type $\tau_y(\theta)$ can also be represented with a different
 217 formulation such as

$$\tau_y(\theta) = \tau_{\max}(j/k_y)(e^{1-j_y(\theta)/k_y}) \quad (20)$$

218 Other formulations can be found in (Smith, 2014). All relevant variables are
 219 depicted in Fig. 1.

220 To assess the accuracy of (16) compared to SCM, a simulation is run in
 221 Chrono (Tasora et al., 2016). The simulation utilizes a single wheel test bed
 222 operating on a sand-like terrain using Chrono's built-in SCM terrain. The test

Table 2: Terrain parameters for sand (Guo, 2016).

Parameter	Value
k_c	1000 (N/m ^{$n+1$})
k_ϕ	1528600 (N/m ^{$n+2$})
n	1.08 (-)
k	0.024 (m)
c	200 (Pa)
ϕ	0.4712 (rad)

Table 3: Wheel states for benchmark simulation.

State	Value
Normal load	2500 (N)
Longitudinal slip	0.2 (-)
Camber angle	0 (rad)
Speed	5.5 (m/s)

bed allows for individual control of the tire’s velocity, load, longitudinal slip, and lateral slip. The terrain properties used in this simulation are representative of sand and given in Table 2. The simulation sweeps the tire through a range of lateral slips with a 1 Hz sine wave. The load, longitudinal slip, camber angle, and linear velocity of the tire are all held at the constant values given in Table 3.

Fig. 2 shows the results of an SCM simulation run in Chrono (orange) and the force predicted by (16) (blue). The term k is assumed to be constant, rather than a function of lateral slip as in (Ishigami et al., 2007). This is to maintain consistency with the SCM formulation used in Chrono. As seen in the figure, the base model of (16) captures the overall trend, at least in the linear region around zero lateral slip, but averages the two distinct curves seen in SCM. This is because the current formulation does not account for the hysteresis effects of varying lateral slip; i.e., the shear deformation of (18) does not account for the shearing resulting from the tire rotation that induces the lateral slip. Note that

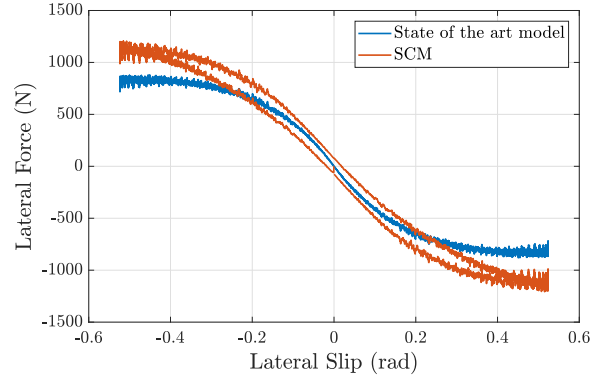


Figure 2: Simulation results for SCM (orange) and model based on (16) (blue). The simulation uses the inputs given in Table 3 and the terrain properties given in Table 2.

in this work the lateral slip is varied by the steering angle applied to the tire.

Recognizing this shortcoming in the state-of-the-art fast terramechanics model, a new surrogate model for SCM is developed in the next section.

2.3. New Surrogate Model

This section presents the new terramechanics model developed as a fast surrogate for the SCM.

The new surrogate is obtained by replacing (17) and (20) with the following expressions, respectively.

$$\tau_y^*(\theta) = \tau_{\max}(1 - e^{-|j_y^*(\theta)|/k_y^*})g_1(v, s, F_z, n) \quad (21)$$

$$\tau_y^*(\theta) = \tau_{\max}(j_y^*(\theta)/k_y^*)(e^{1-j_y^*(\theta)/k_y^*})g_1(v, s, F_z, n) \quad (22)$$

with

$$j_y^*(\theta) = -|r(1-s)(\theta_f - \theta)\tan\beta| + \text{sign}(\beta)(r\sin(\theta)\Delta\delta g_2(v, s, F_z, n)) \quad (23)$$

where $\Delta\delta$ is the change in the steering angle. k_y , a parameter originally de-

248 scribing the shear displacement required to generate peak shear stress, is now
 249 empirically estimated as a function of the wheel states, i.e.,

$$k_y^* = g_3(v, s, F_z, n) \quad (24)$$

250 The lateral force acting on the vehicle is then determined as in (16). It
 251 should be noted that an additional term in (16) is often given representing
 252 the bulldozing force; however, simulations suggest this contribution is minimal
 253 for this application. Additionally, integrating the original nonlinear functions
 254 over the contact patch is a computationally demanding task. Therefore, the
 255 quadratic approximation proposed in (Zhenzhong Jia et al., 2011) is adopted in
 256 the surrogate model. Furthermore, the modifications shown represent the lateral
 257 force acting on the vehicle frame, not the lateral forces in the tire reference frame.

258 Simulations covering the operating range of a notional military AGV are
 259 run to develop the modifying functions $g_1(\cdot) - g_3(\cdot)$. For each slip range of
 260 the clay simulation, as described in the Appendix, the simulations are run at 4
 261 equispaced wheel loads, 5 slips, 5 translational velocities, and 7 sinkage expo-
 262 nents, i.e. 4+5+5+7 samples. It is worth noting, this could potentially cause a
 263 loss of interaction effects between the inputs; however, training with 4+5+5+7
 264 samples rather than 4*5*5*7 samples leads to much faster development time.
 265 Furthermore, to ensure this method leads minimal loss of accuracy from inter-
 266 action effects, over 1,500 independent validation tests were performed for cases
 267 where interacting effects would occur, as described in the following paragraphs.
 268 Finally, other terrain parameters are set to their nominal values, because only
 269 the sinkage exponent is selected as the parameter to be estimated due to the
 270 higher sensitivity of tire forces to sinkage exponent than other parameters (Gal-
 271 lina et al., 2014; Ishigami, 2008). Table 4 shows the range of inputs covered
 272 in the simulations. In these simulations, the inputs are held at constant values

Table 4: Wheel states and terrain ranges for development of $g_1(\cdot) - g_3(\cdot)$

State	Value
Normal load	1000–4000 (N)
Longitudinal slip	-0.9–0.9 (-)
Camber angle	0 (rad)
Speed	2.5–8.5 (m/s)
n	0.4–1.3 (-)

and the lateral slip is varied with a sinusoidal input. Following this, correction factors are determined for τ_y , $j_y(\theta)$, and k_y to match the output of (16) with each of the SCM simulations. Least squares curve fitting is then used to derive the relationship between the correction factors and the simulation inputs of Table 4, resulting in the modification functions $g_1(\cdot) - g_3(\cdot)$. To ensure the model was not subject to overfitting or a loss of accuracy from interaction effects, over 1,500 independent validation simulations were performed as described in the following paragraphs.

Several parameters in the surrogate terramechanics model have distinct effects on the lateral force prediction and can be modified to achieve better agreement with SCM. For illustration purposes, the effects of each input on a sand terrain are shown in Fig. 3-5. The slope of the linear region can be set by modifying k_y with $g_3(\cdot)$, the distance between the two curves can be set by adjusting j_y with $g_2(\cdot)$, and the overall magnitude of the force can be adjusted with $g_1(\cdot)$. The effect of the wheel states on $g_1(\cdot) - g_3(\cdot)$ are as follows. The effect of wheel load can be captured with linear functions for $g_1(\cdot) - g_3(\cdot)$. Increased wheel load tends to increase the magnitude of the lateral force, increase the slope of the linear region, and increase the separation between the top and bottom curves as shown in Fig. 3 for a load 2000N (blue) and 3000N (orange). The effect of longitudinal slip can be modeled by polynomials for $g_1(\cdot) - g_3(\cdot)$. As seen in Fig. 4, for positive slips, lower magnitude longitudinal slips tend to increase the

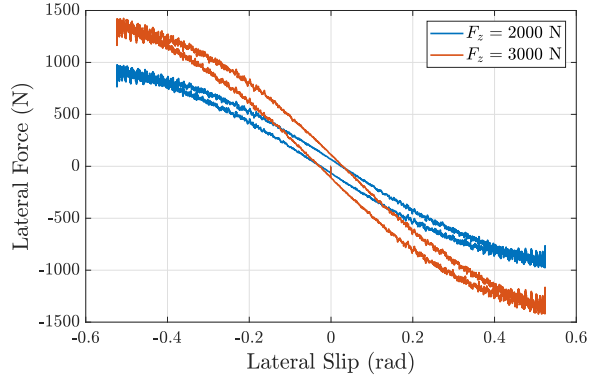


Figure 3: Simulation results for SCM with F_z being 2000 N (blue) and 3000 N (orange). The simulation uses the inputs given in Table 3 and the terrain properties given in Table 2, except for the normal load and slip (-0.5).

294 slope of the linear region, while also causing a larger spread between the top and
 295 bottom curve. Fig. 4 shows these results for a slip of 0.2 (blue), 0.4 (orange),
 296 and 0.6 (yellow). The effect of translational velocity can be captured with a
 297 power function for $g_2(\cdot)$ alone, because it has minimal effect on $g_1(\cdot)$ and $g_3(\cdot)$.
 298 Hence, $g_1(\cdot)$ and $g_3(\cdot)$ do not depend on translational velocity. As seen in Fig.
 299 5, for a speed of 2.5 m/s (blue) and 4.5 m/s (orange), increased translational
 300 velocity tends to have little effect on the slope of the linear region, but reduces
 301 the hysteresis. Example formulations of $g_1(\cdot) - g_3(\cdot)$ for a clay terrain are given
 302 in the Appendix.

303 Once $g_1(\cdot) - g_3(\cdot)$ are determined, over 1,500 independent validation simu-
 304 lations are ran using equispaced samples for the wheel states of Table 4. The
 305 mean absolute error for 1,500 SCM clay simulations is 62.97 N, suggesting the
 306 surrogate model is able to accurately predict the tire forces beyond the training
 307 set and incurs minimal loss of accuracy from the interaction effects of the train-
 308 ing method. The results of the surrogate model are shown in Fig. 6 (blue) while
 309 SCM is shown in (orange), for the same SCM simulation of Fig. 2. Much better
 310 agreement is observed between the surrogate model and the SCM simulation
 311 compared to Fig. 2. It should also be noted that the surrogate model runs in

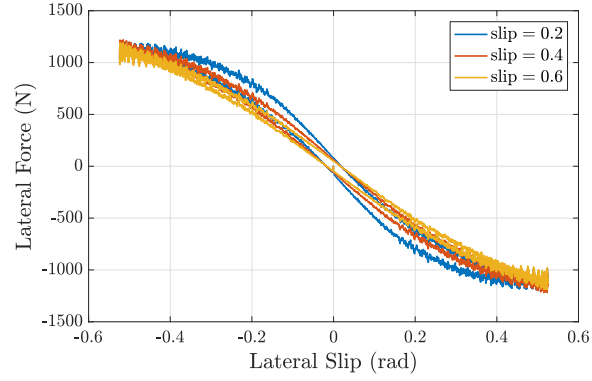


Figure 4: Simulation results for SCM with slip being 0.2 (blue), 0.4 (orange), 0.6 (yellow). The simulation uses the inputs given in Table 3 and the terrain properties given in Table 2, except for the slip.

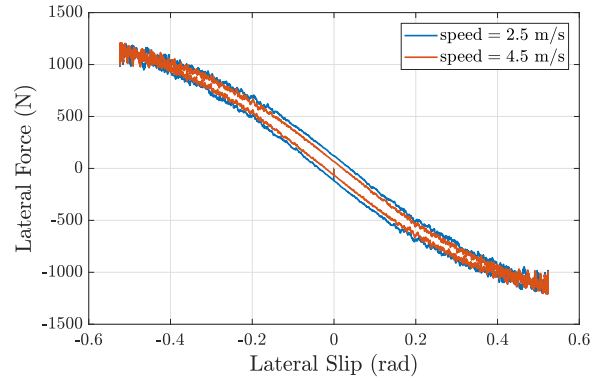


Figure 5: Simulation results for SCM with speed being 2.5 m/s (blue) and 4.5 m/s (orange). The simulation uses the inputs given in Table 3 and the terrain properties given in Table 2, except for the speed and slip (0.5).

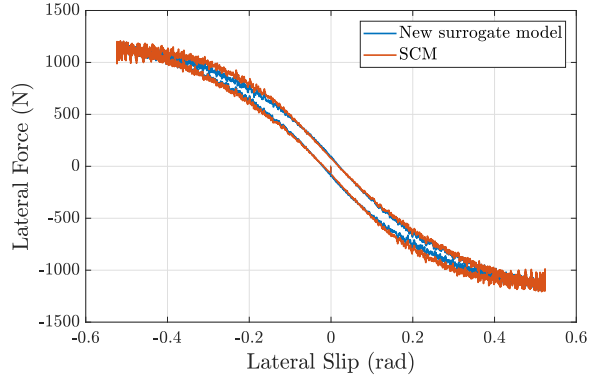


Figure 6: Simulation results for SCM (blue) and the new surrogate model (orange) for the inputs given in Table 3 and the terrain properties given in Table 2.

200-400 μs , which is an order of magnitude more efficient than what is reported
for SCM and more suitable for online terrain estimation.

3. Vehicle Models

Two vehicle models are employed in this work; a 11 DoF model to represent
the plant and a 3 DoF bicycle model to be used as part of the estimator. This
section describes these models.

3.1. Plant Model

To represent the actual vehicle in the simulation-based validation of the
proposed surrogate model and terrain estimator, the Chrono software is utilized
to simulate the dynamics of a notional military vehicle, specifically a Polaris
MRZR 4, as well as to implement the SCM (Tasora et al., 2016). For the
purposes of this work, the vehicle is modeled with a double wishbone suspension,
rack-pinion steering, 4 wheel drive, and a simple powertrain without a torque
converter or transmission. This results in a 11 DoF vehicle model that is then
combined with the SCM as the tire-terrain interaction model. Fig.7 depicts the
plant operating on a sand SCM terrain in Chrono. The data received from the

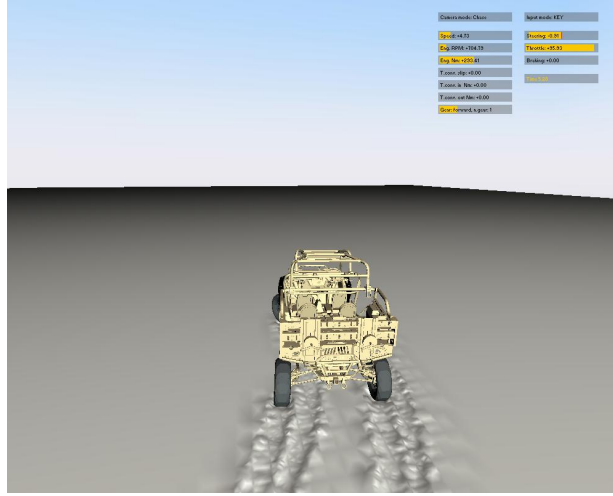


Figure 7: Plant operating on sand SCM terrain in Chrono.

Table 5: Measurement standard deviations used for sensor simulation.

State	Noise (σ)
x	1.2 (m)
y	1.2 (m)
ψ	0.0175 (rad)
u	0.25 (m/s)
v	0.25 (m/s)
ω_z	0.0175 (rad/s)

328 plant is then corrupted with Gaussian noise and serves as the measurement y_k
 329 in (29) in Sec. 4. Table 5 lists the standard deviations used in the noise model
 330 for each state. Actual sensors typically offer lower noise levels; hence the chosen
 331 standard deviations represent a worse-case scenario to test the ability of the
 332 estimator (Ryu et al., 2002).

333 3.2. Bicycle Model

334 As part of the terrain estimation process that is detailed in Sec. 4, a vehicle
 335 model is needed to predict future vehicle states based on the tire forces from
 336 the surrogate model. For this work, a 3 DoF bicycle model with forward Euler

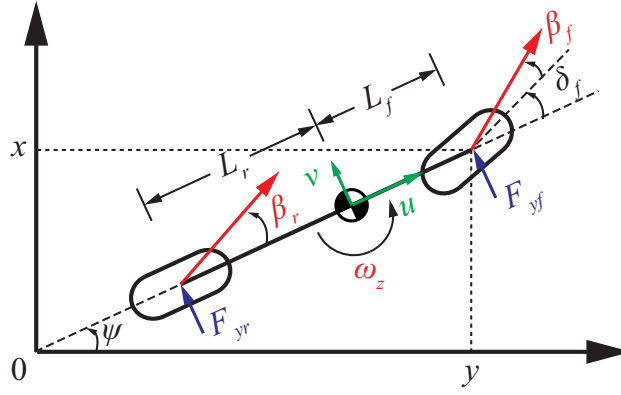


Figure 8: Bicycle model schematic (Liu et al., 2017)

337 integration is adopted, as it provides a proper level of fidelity while maintaining
 338 enough simplicity for short-horizon predictions (Liu et al., 2016). Fig. 8 depicts
 339 the bicycle model schematic. Mathematically, the bicycle model takes on the
 340 following form:

$$\dot{z}_b = \begin{bmatrix} u \cos \psi - (v + L_f \omega_z) \sin \psi \\ u \sin \psi + (v + L_f \omega_z) \cos \psi \\ w_z \\ a_x \\ (F_{yf} + F_{yr})/M_t - u \omega_z \\ (F_{yf} L_f - F_{yr} L_r)/I_{zz} \end{bmatrix} \quad (25)$$

341 where the state vector, z_b , is defined as

$$z_b := \begin{bmatrix} x \\ y \\ \psi \\ u \\ v \\ \omega_z \end{bmatrix} = \begin{bmatrix} \text{global } x \text{ position of front axle} \\ \text{global } y \text{ position of front axle} \\ \text{yaw angle} \\ \text{longitudinal velocity} \\ \text{lateral velocity} \\ \text{yaw rate} \end{bmatrix} \quad (26)$$

342 with M_t being the vehicle mass, I_{zz} being the vehicle's yaw moment of inertia,
 343 and L_f and L_r being the distance from the vehicle's center of gravity to the front
 344 and rear axles, respectively. Finally, F_{yf} and F_{yr} are the lateral forces generated
 345 from the front and rear tires acting on the vehicle body, as obtained from the
 346 terramechanics model.

347 4. Terrain Estimation

348 The terrain parameter to be estimated is chosen as the sinkage exponent n ,
 349 because it has been shown to be the dominant parameter (Gallina et al., 2014).
 350 All other terrain parameters are assumed to be some nominal values based on
 351 the specific terrain type, which can be determined from terrain classification
 352 algorithms such as the ones described in (Iagnemma, 2006; Weiss et al., 2008).

353 To estimate the unknown terrain parameter n , it is appended to the 3 DoF
 354 bicycle model in (25) with trivial dynamics. Here n is given as a 2x1 vector
 355 to account for the front and rear tires. This is to mitigate the influence of
 356 unmodeled multipass effects and in the case of a discrete terrain change where
 357 the front tire and rear tire may operate on different terrains. The augmented

state vector and state dynamics are given as

$$z := \begin{bmatrix} z_b \\ n \end{bmatrix}, \quad \dot{z} = \begin{bmatrix} \dot{z}_b \\ 0 \end{bmatrix} \quad (27)$$

Given the measurements of the vehicle states in (26), the augmented dynamics are utilized in an unscented Kalman filter (UKF) to estimate the augmented state vector in (27) including the sinkage exponent. It is worth noting that many other algorithms are available in the literature for nonlinear parameter estimation, including, but not limited to, extended Kalman filters, transitional Markov Chain Monte Carlo algorithms, and particle filters. Among these options the UKF is preferred in this work, because preliminary explorations suggest that the UKF offers a good balance between accuracy and computational speed for this application.

The UKF is composed of two general steps; a time update step and a measurement update step. Assume that a system is given in discrete time as:

$$z_{k+1} = F(z_k, \rho_k) \quad (28)$$

370

$$y_k = H(z_k, \gamma_k) \quad (29)$$

where z is the state, y is the observation, and ρ and γ are the process and observation noise, respectively. The functions $F(\cdot)$ and $H(\cdot)$ are nonlinear functions describing the dynamics and outputs. In this application, z takes the form of the state vector in (27) and $F(\cdot)$ is obtained by discretizing the state equation in (27) using the forward Euler method. $H(\cdot)$ is given as the state vector in (26).

First, a set of $2L + 1$ sigma points are created to capture the statistical distribution of the states, where L is the dimension of the state vector z . The

379 sigma points are determined as follows:

$$Z_{k-1} = [\hat{z} \quad \hat{z} \pm (\sqrt{(L + \lambda)P_z})_i] \quad (30)$$

380 where $\hat{z}$ is the mean value of z . λ is a scaling parameter given as:

$$\lambda = \alpha^2(L + \kappa) - L \quad (31)$$

381 where α is a tunable scaling parameter that typically takes a value between 0
 382 and 1, κ is another scaling parameter that is typically set to 0. At the time
 383 update step, the sigma points are propagated through the original nonlinear
 384 system as:

$$Z_{k|k-1}^z = F(Z_{k-1}^z, Z_{k-1}^v) \quad (32)$$

385 The following weights are then calculated

$$W_0^m = \lambda/(L + \lambda) \quad (33)$$

386

$$W_0^c = \lambda/(L + \lambda) + (1 - \alpha^2 + \zeta) \quad (34)$$

387

$$W_i^{m,c} = 1/(2(L + \lambda)) \quad (35)$$

388 where ζ is set to 2 for Gaussian distributions. The statistics of the time update
 389 step are then given by:

$$\hat{z}_k^- = \sum_{i=0}^{2L} W_i^m Z_{i,k|k-1}^z \quad (36)$$

390

$$P_k^- = \sum_{i=0}^{2L} W_i^c (Z_{i,k|k-1}^z - \hat{z}_k^-)(Z_{i,k|k-1}^z - \hat{z}_k^-)^T \quad (37)$$

391

$$Y_{k|k-1} = H(Z_{k|k-1}^z, Z_{k-1}^n) \quad (38)$$

$$\hat{y}_k^- = \sum_{i=0}^{2L} W_i^m Y_{i,k|k-1} \quad (39)$$

Finally, the measurement update step is given by the following set of equations:

$$P_{\hat{y}_k \hat{y}_k} = \sum_{i=0}^{2L} W_i^c (Y_{i,k|k-1} - \hat{y}_k^-)(Y_{i,k|k-1} - \hat{y}_k^-)^T \quad (40)$$

$$P_{z_k y_k} = \sum_{i=0}^{2L} W_i^c (Z_{i,k|k-1} - \hat{z}_k^-)(Y_{i,k|k-1} - \hat{y}_k^-)^T \quad (41)$$

$$K = P_{z_k y_k} P_{\hat{y}_k \hat{y}_k}^{-1} \quad (42)$$

$$\hat{z}_k = \hat{z}_k^- + K(y_k - \hat{y}_k^-) \quad (43)$$

$$P_k = P_k^- - K P_{\hat{y}_k \hat{y}_k} K^T \quad (44)$$

The above process is a summary of the algorithm given in (Wan and Van Der Merwe, 2000). Intuitively, the process works by merging model-based predictions of the states with their measurements from the plant by exploiting the uncertainties associated with each to determine the best estimates of the states. For further discussion of UKF and details of its implementation, the reader is referred to (Wan and Van Der Merwe, 2000; Kolås et al., 2009). It also worth noting that while this work only focuses on estimating the dominant parameter n , other terrain parameters could be estimated simultaneously, as well. However, this would incur additional computational costs as the state space dimension increases, thus increasing the number of sigma points necessary in the UKF.

5. Simulation Setup

Simulations are performed utilizing Chrono's SCM deformable terrain and the developed AGV model in Sec. 3.1. Two terrains are considered including

Table 6: Terrain parameters for simulated terrains (Smith, 2014).

Parameter	Sandy Loam	Clay
k_c	5300 (N/m ⁿ⁺¹)	13200 (N/m ⁿ⁺¹)
k_ϕ	1515000 (N/m ⁿ⁺²)	692200 (N/m ⁿ⁺²)
n	0.7 (-)	0.5 (-)
k	0.025 (m)	0.01 (m)
c	1700 (Pa)	4140 (Pa)
ϕ	0.5061 (rad)	0.2269 (rad)

sandy loam and clay. Relevant terrain parameters are given in Table 6. The simulation subjects the AGV to sinusoidal steering commands, steering fully to the left and right over a three second period. The throttle is also varied with a sinusoidal command such that varying speeds are achieved. No braking command is given. The applied steering and speed profiles for the clay simulation are shown in Fig. 9. Throttle and steering commands of the same frequency are given in the sandy loam simulation, as well. Two remarks are in order. First, as seen in Fig. 9, no requirement on constraining the vehicle to low speeds (on the order of 10 cm/s) is made, which is in contrast to previous efforts (Iagnemma et al., 2004). This enables enhanced mobility, which is critical for military applications. Second, a sinusoidal steering input is selected to induce lateral dynamics for the vehicle. Since the bicycle model only utilizes the lateral forces acting on the vehicle, it is critical for the estimation that the vehicle operates in such a way that lateral dynamics are induced. Otherwise, the lack of information on the lateral dynamics leads to parameter variations having negligible effects on the output of the bicycle model. In other words, if F_{yf} and F_{yr} are zero, it is not possible to estimate terrain parameters based on lateral forces. If estimation is required in such a scenario, additional measurement should be exploited.

In all simulations, the simulation time step is set to 2 ms in Chrono. The purpose of these simulations is to determine the estimation algorithm’s accuracy and utility under different terrain conditions. Once the simulations are com-

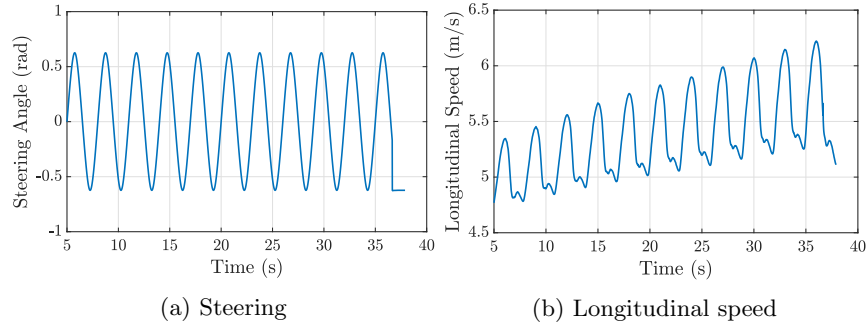


Figure 9: Steering and velocity profiles used in simulation.

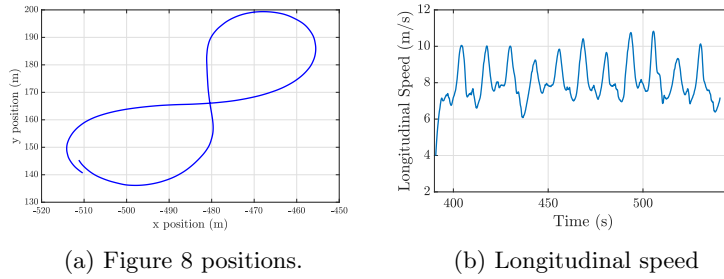


Figure 10: Position and velocity profiles from the experiment on grass.

434 plete, noise is added to the outputs to simulate sensors, as discussed in Sec. 3.1.
 435 The estimator is then run at a 12 ms time step and the simulated measurements
 436 are received at every 24 ms. The terrain parameter n is initialized with a value
 437 off of the true terrain parameter used in the plant simulation. The remaining
 438 terrain parameters are set to their true values given in Table 6. Note that the
 439 true values are used here only to assess how closely the algorithm can converge
 440 to the true sinkage exponent.

441 **6. Experimental Setup**

442 Experiments were performed with a Polaris MRZR 4, example shown in
 443 Fig. 11, equipped with sensors to measure the relevant states required for the
 444 measurement update of the UKF as described in Sec. 4. Actuators were con-



Figure 11: Example Polaris MRZR 4 (dvidshub, accessed January 30, 2020).

Table 7: Terrain parameters for grass field (Okello et al., 1998).

Parameter	Value
k_c	16000 (N/m ⁿ⁺¹)
k_ϕ	1180000 (N/m ⁿ⁺²)
n	0.26 (—)
k_w	0.046 (m)
c	24000 (Pa)
ϕ	0.75 (rad)

445 trolled with TORC ByWire, Actus ERC 722 for throttle and braking, and Allied
 446 Motion GLOBE POW-R STEER Electric Power Assisted Steering (EPAS) for
 447 steering. A Robotic Research RR-N140 navigation system was used for high
 448 accuracy localization which is composed of Global Navigation Satellite System
 449 (GNSS)/Global Positioning System (GPS) as well as an Inertial Measurement
 450 Unit (IMU). In addition, Honeywell SNDH-T4L-G01 were used for wheel en-
 451 coders. Sensor noise values are given in Table 5. Measurements were received
 452 by the UKF at a frequency of 25 Hz and the bicycle model predictions were
 453 made at a frequency of 50 Hz.

454 The vehicle was subjected to a series of Figure 8 maneuvers as exemplified in
 455 Fig. 10 (a) with a longitudinal velocity profile given in 10 (b). This maneuver
 456 was selected to induce lateral dynamics. The location of the maneuver was a wet
 457 open grass field. While the exact terrain parameters of the field are unknown,
 458 the literature provides several parameter sets for various grass terrains (Okello

et al., 1998). The terrain parameters found in the literature for a terrain most comparable to the test site were chosen as shown in Table 7 due to the moisture content and surface type.

7. Results and Discussion

7.1. Simulation Results

In this section the performance of the terrain estimator is evaluated. The performance is evaluated from two different points of view: (1) the accuracy of the estimated sinkage exponent n , and (2) the accuracy of the predicted state trajectories of the vehicle. The former assesses the algorithm's ability to find the true sinkage exponent, whereas the latter assesses the utility of estimating the sinkage exponent in the larger picture of predicting the future states of the vehicle. Note that if the assumed nominal values for the non-estimated parameters are not representative of the true terrain type, then the estimator may not necessarily converge to the true terrain parameter, as it will attempt to find a value that compensates for the errors in the non-estimated values and achieves the best prediction capability of the vehicle model. For the ultimate aim of more accurately predicting the future mobility capabilities of the vehicle, the second evaluation criterion is the more relevant one.

Table 8 displays the initial guess of the value of the sinkage exponent n , its converged estimate by the algorithm, and the error associated with the estimated parameter. The initial terrain parameter for sandy loam is chosen to be representative of Buchele (Michigan) sandy loam and the initial terrain parameter for clay is chosen to be representative of Thailand clay (Wong, 2001). On both terrains, the percent error in the estimated terrain parameter is less than 4%, where the estimated value is taken to be the final value by the end of the simulation. Fig. 12 shows the estimated terrain parameters for the two

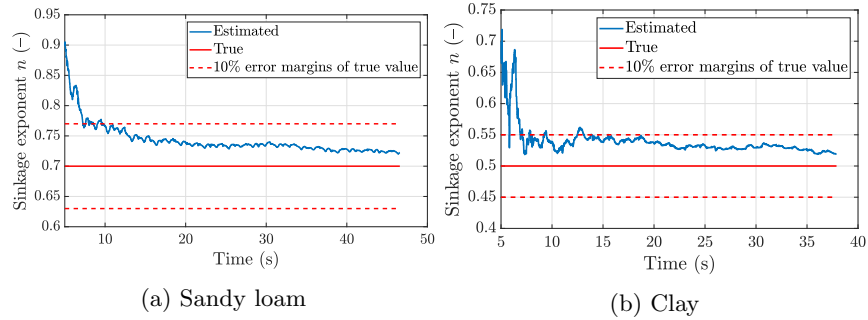


Figure 12: Simulated sinkage exponent estimation results.

Table 8: Initial guess, estimated value, and estimation errors of the sinkage exponent n for simulated terrains.

Terrain	Initial guess	True val.	Converged val.	% error
Sandy loam	0.9	0.7	0.722	3.1%
Clay	0.7	0.5	0.519	3.8%

considered terrains as time evolves. The differences between the converged and true terrain values can be due to model discrepancies between the high fidelity Chrono model and the 3 DoF bicycle model along with discrepancies arising from the reduced order terramechanics model. Nevertheless, the estimator converges within 10% of the estimated parameter within 5 seconds for both cases.

The peak computation time of the estimator is 10.5 ms and 7.5 ms for clay and sandy loam simulations respectively, thus demonstrating the potential to achieve real-time estimation. The platform running this estimation consists of 16 GB Memory and a single core 3 GHz Intel Core i7 processor.

While estimating the terrain properties accurately is a worthy goal in and of itself, it is more of interest to evaluate to what extent the estimations can improve the predictive capability of the bicycle model as motivated above. To accomplish this second evaluation, the bicycle model, with the terramechanics model parameterized by either the initial guess or the converged terrain parameter, is used to predict the vehicle states approximately 0.5, 2.5, and 5.0 seconds

Table 9: Mean squared error over entire simulation with varying prediction horizons for clay using estimated terrain parameter ($n = 0.519$) and initial guess ($n = 0.7$).

Time horizon	0.5 (s)		2.5 (s)		5 (s)	
State	n=0.519	n=0.7	n=0.519	n=0.7	n=0.519	n=0.7
x (m)	0.0034	0.0025	0.037	0.01	0.075	0.078
y (m)	0.0035	0.0051	0.022	0.15	0.12	0.34
ψ (rad)	3.17e-05	1.8e-04	2.45e-04	0.0089	9.9e-04	0.0115
u (m/s)	6.46e-05	6.46e-05	1.33e-04	1.33e-04	1.2e-04	1.2e-04
v (m/s)	0.0027	0.013	0.0047	0.15	0.005	0.28
ω_z (rad/s)	6.2e-04	0.004 (rad/s)	9.01e-04	0.023	9.05e-04	0.04

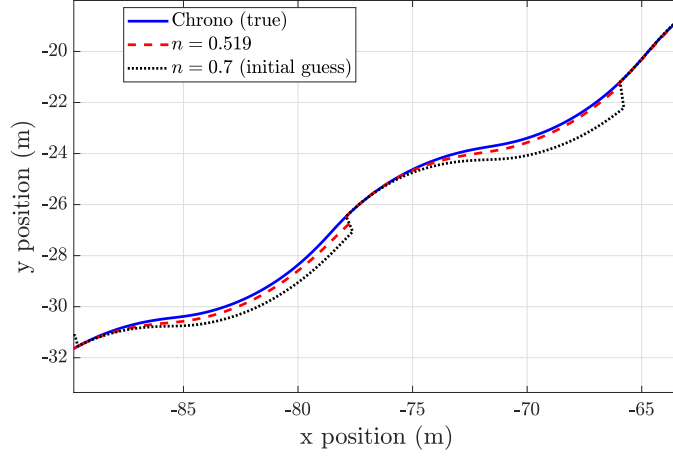


Figure 13: Simulated vehicle positions for AGV operating on clay. True vehicle positions from Chrono (blue solid line), bicycle model parameterized by $n = 0.519$ (red dashed line), and bicycle model parameterized by initial terrain guess $n = 0.7$ (black dotted line).

500 into the future for the clay case. After this time the vehicle states are reset
501 to the true values received from Chrono. As such, this procedure mimics the
502 operational procedure of a model predictive control approach, where a reced-
503 ing finite horizon optimal control problem is solved periodically with updated
504 information available from sensors (Liu et al., 2017).

505 Table 9 depicts the mean squared errors (MSE) of the state estimates given
506 by the 3 DoF bicycle model for both the case when the initial guess for the
507 sinkage exponent for clay is used and the case when the converged estimate

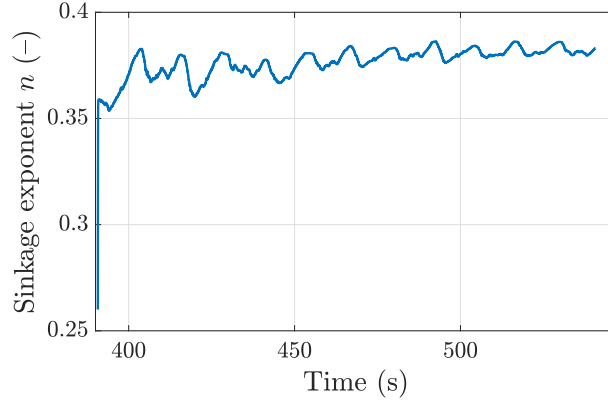


Figure 14: Sinkage exponent estimation results for the experiment on grass.

is used over the entire 32.89 s simulation. The model parameterized by the estimated terrain property yields significantly better predictions, especially at larger time horizons with order of magnitude reductions in MSE. Fig. 13 shows a portion of the simulation, using the ~ 2.5 s time horizon and depicting the true vehicle positions from the plant (blue solid line), and the predicted positions using the bicycle model with the initial guess of the sinkage exponent (black dotted line) and with the converged sinkage exponent (red dashed line). As can be seen, the converged value yields much more accurate predictions, thus demonstrating the ability of the estimator to significantly improve prediction fidelity. Similar results are also observed for the other terrains, but they are not reported here due to space limitations. This improvement in turn could lead to better performance in model predictive controllers, which is subject to future research.

7.2. Experimental Results

Fig. 14 shows the estimated sinkage exponent for the experimental tests on grass described in Sec. 6. Here, the initial guess of the sinkage exponent corresponds to that of Table 7 and all other terrain parameters are set to nominal

Table 10: Mean squared error over entire experiment with 2.5 second prediction horizon for the experiment on grass.

State	n=0.39 (estimated value)	n=0.26 (initial guess)
x	0.3132 (m)	1.7305 (m)
y	0.4378 (m)	1.6483 (m)
ψ	0.0117 (rad)	0.0209 (rad)
u	1.8747e-04 (m/s)	1.8747e-04 (m/s)
v	0.1740 (m/s)	0.3989 (m/s)
ω_z	0.0100 (rad/s)	0.0204 (rad/s)

values given in Table 7. While the true sinkage exponent of the grass test site is unknown, the converged value of 0.39 is within the range reported for grass of 0.26-0.7 as given in (Okello et al., 1998).

Table 10 displays the MSE of the state predictions of a 3 DoF bicycle model parameterized by the initial sinkage exponent guess ($n = 0.26$) and the converged estimate ($n = 0.39$) as compared to measurements on the experimental vehicle over a 2.5 s horizon. As can be seen, the model parameterized by the estimate significantly reduces the MSE for all states, demonstrating the utility of estimator. Furthermore, this result provides confidence in the estimated parameter value despite uncertainty in its true value. This is demonstrated by Fig. 15, which plots the normalized sum squared error (SSE) over the entire experiment between a bicycle model prediction (one time step into the future) as compared to state measurements from the vehicle as a function of the sinkage exponent. As can be seen, the sinkage exponent that minimizes this error is approximately 0.395, which is within 1.5% of the estimated parameter value. Thus the experimental results demonstrate the ability of the estimator to improve the prediction capabilities of the bicycle model, which could prove beneficial in model predictive navigation strategies.

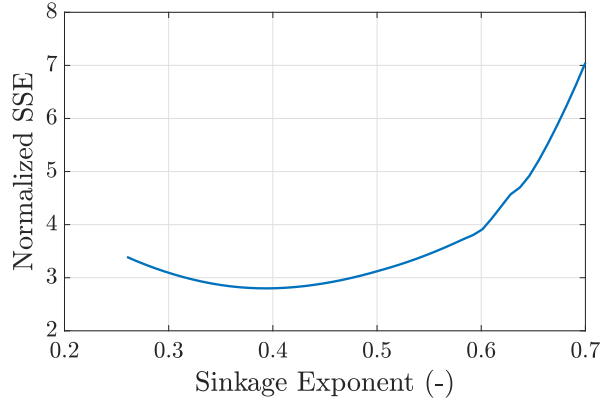


Figure 15: Normalized sum squared error predictions for various sinkage exponents for the experiment on grass.

543 7.3. Remarks

544 While the results of the estimator are promising, there are two limiting
545 assumptions of the proposed scheme. The first is that the estimator assumes the
546 terrain is homogeneous. This is common among the approaches reported in the
547 literature (Iagnemma et al., 2004). However, in reality, terrain parameters may
548 be changing and evaluating the performance of the estimator in this scenario is
549 subject to future work. The second limiting assumption is that SCM is treated
550 as the ground truth and the surrogate terramechanics model is parameterized
551 accordingly. The reality may be different than SCM. For example, SCM was
552 originally developed for rover applications which tend to occur at lower speeds.
553 At high speeds, inertial effects may impact model fidelity. However, in that
554 case, a similar procedure in developing the surrogate model could potentially
555 be used by replacing the SCM simulations with an experimental single wheel
556 test bed. While the experimental results presented herein suggest the estimator
557 is able to improve the prediction capability of the bicycle model despite this
558 second limitation, it is still of interest to experimentally validate the developed
559 terramechanics model, which is subject to future work.

560 8. Conclusion

561 This paper considers AGVs operating on deformable terrains with unknown
562 terrain properties and develops a novel terrain estimation framework towards
563 increasing the terrain-awareness of the AGV. In particular, the novelty of the
564 framework is the development of a new surrogate terramechanics model for SCM
565 and its use in conjunction with a bicycle model in a UKF. The results suggest
566 that this new framework can estimate the dominant terrain parameter, namely
567 the sinkage exponent, with high accuracy and high computational efficiency. It is
568 therefore concluded that the framework is an important step towards achieving
569 a good balance between estimation accuracy and computational speed. The
570 results also show that the increase in the accuracy of the terrain parameter
571 due to the developed estimation framework leads to a significant increase in
572 the predictive accuracy of the bicycle model, especially for longer prediction
573 time horizons. It is therefore concluded that the proposed framework could
574 be useful to increase the performance of AGVs when they are controlled with
575 model predictive schemes.

576 Future work includes evaluating the estimator on varying terrain conditions.
577 The need for and ability of estimating multiple terrain parameters also needs
578 to be investigated. It is also of interest to perform experimental validation of
579 the developed surrogate terramechanics model, and investigate the utility of the
580 estimator in a model predictive control framework for terrain-aware autonomous
581 navigation.

582 Appendix

583 The following depicts the formulas for $g_1 - g_3$ for a clay terrain. In this work
584 the equations for $g_1 - g_3$ are determined for four separate slip ranges for better

585 agreement. The slip ranges are:

$$0.16 \leq s \quad (45)$$

586

$$0 \leq s < 0.16 \quad (46)$$

587

$$-0.157 < s < 0 \quad (47)$$

588

$$s \leq -0.157 \quad (48)$$

589 The dependencies for $g_1 - g_3$ on each input is determined through curve fitting
 590 to simulation data as discussed in Sec. 2.3. The below equations are valid for
 591 the slip range of $0 \leq s < 0.16$. For the lower curve of Fig. 6, the equations are
 592 given as:

$$g_1 = g_{1_n} g_{1_s} \quad (49)$$

593

$$g_2 = g_{2_n} g_{2_s} g_{2_{F_z}} g_{2_v} \quad (50)$$

594

$$g_3 = g_{3_n} g_{3_s} g_{3_{F_z}} \quad (51)$$

595 where

$$\begin{aligned} g_{1_n} = & 128.3n^5 - 415.4n^4 + 523.3n^3 \\ & - 320n^2 + 95.07n - 9.942 \end{aligned} \quad (52)$$

596

$$\begin{aligned} g_{1_s} = & 563.5s^4 - 107.9s^3 - 4.848s^2 \\ & + 1.761s + 1.024 \end{aligned} \quad (53)$$

597

$$g_{2_n} = \max(1.8n - 1.08, 0) \quad (54)$$

598

$$\begin{aligned} g_{2_s} = & 1.367 \times 10^6 s^4 - 3.71 \times 10^5 s^3 \\ & + 2.809 \times 10^4 s^2 - 545.7s + 70 \end{aligned} \quad (55)$$

$$g_{2_{F_z}} = (0.0001F_z + 0.7) \quad (56)$$

$$g_{2_v} = (4.908v^{-0.9295}) \quad (57)$$

$$g_{3_n} = -0.1235n^2 + 0.7287n + 0.08425 \quad (58)$$

$$g_{3_s} = -309.5s^4 + 90.48s^3 - 8.983s^2 + 0.2631s + 0.086 \quad (59)$$

$$g_{3_{F_z}} = (8.3 \times 10^{-5})F_z + 0.76 \quad (60)$$

The same process yields the following equations for the upper curve of Fig.

6

$$g_1 = g_{1_s} \quad (61)$$

$$g_2 = g_{2_n} g_{2_s} g_{2_{F_z}} g_{2_v} \quad (62)$$

$$g_3 = g_{3_n} g_{3_s} g_{3_{F_z}} \quad (63)$$

where

$$g_{1_s} = 1.16(1913s^4 - 520.6s^3 + 49.57s^2 - 1.204s + 1.024) \quad (64)$$

$$g_{2_n} = \max(1.8n - 1.08, 0) \quad (65)$$

$$g_{2_s} = 1.367 \times 10^6 s^4 - 3.71 \times 10^5 s^3 + 2.809 \times 10^4 s^2 - 545.7s + 70 \quad (66)$$

$$g_{2_{F_z}} = (0.0001F_z + 0.7) \quad (67)$$

$$g_{2_v} = (4.908v^{-0.9295}) \quad (68)$$

$$g_{3_n} = -0.22 \quad (69)$$

$$g_{3_s} = 1609s^4 - 529s^3 + 58.88s^2 - 2.467s + 0.082 \quad (70)$$

$$g_{3_{F_z}} = (8.3 \times 10^{-5})F_z + 0.76 \quad (71)$$

The equations for other slip ranges can easily be determined by repeating the curve fitting process on data in those ranges. Furthermore, it should be noted that these particular equations are only valid for the specific tire under consideration. Should these be used for a different tire, for example of different radius, the equations are no longer valid and the process would need to be repeated.

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